



EventMad Project

Open Data & Knowledge Graph

Group 07

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Goals

02 Methodology

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Future Work



Introduction

2 Open databases provided by the Madrid City Council on cultural events in the city and the installations

- Activities, performances and exhibitions
- Locations
- Schedules
- Description

Tool that provides quick access to Madrid's cultural offerings, allowing users to tailor their searches based on personal interests and enhancing the cultural experience in the city



Goals

1

Facilitate Access to Cultural Information

2

Innovate Accessibility to Open Data

3

Application of Knowledge in Data Analysis and Visualization



Methodology

1

Create Ontology

Draw.io and CHOWLK



2

Data Preprocessing

OpenRefine



3

Mappings

Github, RMLmapper, SPARQL



4

Data Linking

OpenRefine



5

Data Publishing

Helio, graphDB



6

Application

python, Streamlit



1

Create Ontology

Draw.io and CHOWLK



Data Preprocessing

OpenRefine
OpenRefine

Mappings

Github, RMLmapper,
SPARQL



Data Linking

OpenRefine
OpenRefine

Data Publishing

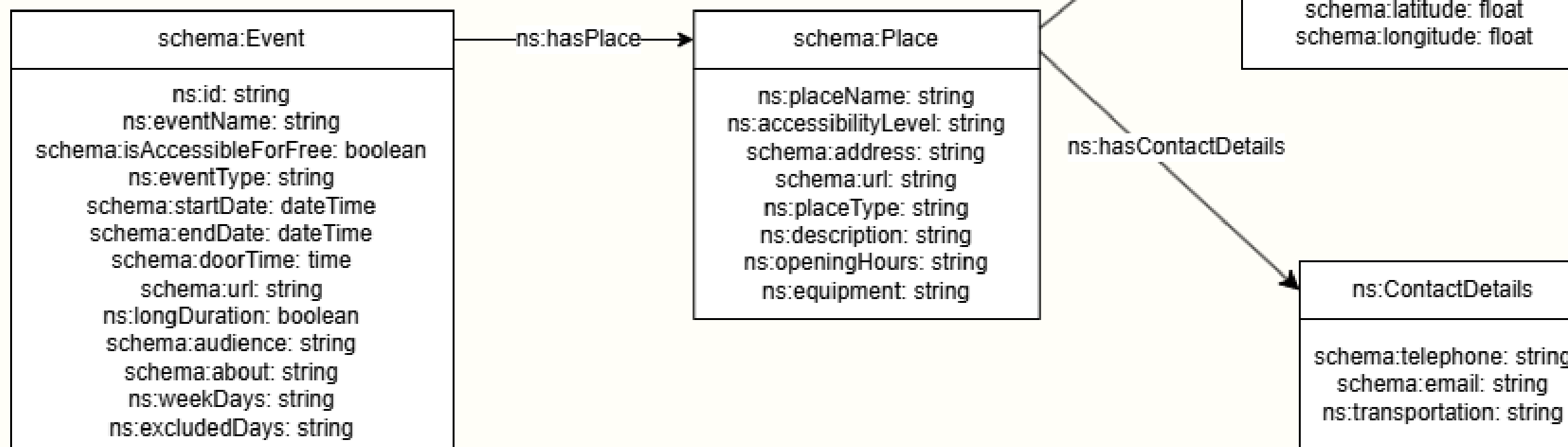
Helio, graphDB
H GraphDB

Application

python, Streamlit
Python Streamlit

ns: <http://madrid.eventos.com/>
schema: <https://schema.org/>

dc:creator Group07
owl:versionInfo: 0.0.1
dc:title MadridEvents Ontology



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Data Preprocessing

OpenRefine
OpenRefine

Mappings

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Data Linking

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python Streamlit

1389 rows

Extensions Wikibase

Show as: rows records Show: 5 10 25 50 100 500 1000 rows « first < previous 1 next > last »

▼ All	▼ ID-EVENTO	▼ TITULO	▼ PRECIO	▼ GRATUITO	▼ LARGA-DURACION	▼ DIAS-SEMANA	▼ DIAS-EXCLUIDOS	▼ FECHA	▼ FECHA-FIN
☆	2.	12548630	17.ª edición Mi primer festival de cine Talleres	0	0			2024-11-23 00:00:00.0	2024-11-23 23:59:00.0
☆	3.	12513819	17ª edición Mi primer festival de cine	0	1	D		2024-11-10 00:00:00.0	2024-11-24 23:59:00.0
☆	4.	12548497	17ª edición Mi primer festival de cine Talleres	0	0			2024-11-16 00:00:00.0	2024-11-23 23:59:00.0
☆	5.	12470350	30 AÑOS DEL PERIÓDICO TETUÁN 30 DÍAS	1	1	L,M,X,J,V,S,D		2024-09-26 00:00:00.0	2024-10-25 23:59:00.0

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Mappings

Github, RMLmapper,
SPARQL



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python Streamlit

RML

```
@prefix rr: <http://www.w3.org/ns/r2rml#> .
@prefix rml: <http://semweb.mmlab.be/ns/rml#> .
@prefix ql: <http://semweb.mmlab.be/ns/ql#> .
@prefix schema: <http://schema.org/> .
@prefix ns: <http://madrid.events.com/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@base <http://madrid.events.com/ns#> .
@prefix xml: <http://www.w3.org/XML/1998/namespace> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .

### TriplesMap for Event ###
<#TriplesMapEvent> a rr:TriplesMap;
  rml:logicalSource [
    rml:source "Dataset-Updated-events-new.csv";
    rml:referenceFormulation ql:CSV
  ];
  rr:subjectMap [
    rr:template "http://madrid.events.com/event/{ID-EVENTO}";
    rr:class schema:Event
  ];

  rr:predicateObjectMap [
    rr:predicate ns:id;
    rr:objectMap [ rml:reference "ID-EVENTO"; rr:datatype xsd:string ]
```

RDF

```
<http://madrid.events.com/event/10590577> <http://madrid.events.com/eventName> "Conferencia: Pierre-auguste Renoir" .
<http://madrid.events.com/event/10590577> <http://madrid.events.com/eventType> "ConferenciasColoquios" .
<http://madrid.events.com/event/10590577> <http://madrid.events.com/hasPlace> <http://madrid.events.com/installation/Centro%20Cultu
20Salmer%C3%B3n%20%28Chamart%C3%ADn%29> .
<http://madrid.events.com/event/10590577> <http://madrid.events.com/id> "10590577" .
<http://madrid.events.com/event/10590577> <http://madrid.events.com/longDuration> "false"^^<http://www.w3.org/2001/XMLSchema#boolean> .
<http://madrid.events.com/event/10590577> <http://schema.org/about> "Ciclo: El Impresionismo. Pierre-Auguste Renoir, a diferencia del resto de los
impresionistas, fue pintor de la figura humana. Su pasión por los maestros del pasado y una especial atención a la mujer y a la sensualidad nos regalar
algunas de las imágenes más bellas del Impresionismo. Ponente: Belén G. Naharro." .
<http://madrid.events.com/event/10590577> <http://schema.org/audience> "Sin información" .
<http://madrid.events.com/event/10590577> <http://schema.org/doorTime> "19:00:00"^^<http://www.w3.org/2001/XMLSchema#time> .
<http://madrid.events.com/event/10590577> <http://schema.org/endDate> "2024-10-30T23:59:00Z"^^<http://www.w3.org/2001/XMLSchema#dateTime> .
```

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Data Linking

OpenRefine

 OpenRefine

Data Publishing

Helio, graphDB



graphDB

Application

python, Streamlit



Streamlit

☐ BARRIO	☐ WIKIDATA_BARRIO	☐ DISTRITO	☐ WIKIDATA_DISTRITO
Cortes	https://wikidata.org/entity/Q2535637 edit	Centro	https://wikidata.org/entity/Q1763376
Cortes	https://wikidata.org/entity/Q2535637	Centro	https://wikidata.org/entity/Q1763376

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Data Publishing

Helio, graphDB



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GraphDB

Import

Explore

Graphs overview

Class hierarchy

Class relationships

Visual graph

Similarity

SPARQL

Monitor

Setup

Lab

events

Source: <http://madrid.events.com/graph/events>

subject

predicate

object

context

all

Explicit only

Show Blank Nodes

Download as

Visual graph

Filter query results

Compact view

Hide row numbers

	subject	predicate	object	context
1	http://madrid.events.com/event/10590577	rdf:type	http://schema.org/Event	http://madrid.events.com/graph/events
2	http://madrid.events.com/event/10606981	rdf:type	http://schema.org/Event	http://madrid.events.com/graph/events
3	http://madrid.events.com/event/10633173	rdf:type	http://schema.org/Event	http://madrid.events.com/graph/events
4	http://madrid.events.com/event/10634056	rdf:type	http://schema.org/Event	http://madrid.events.com/graph/events
5	http://madrid.events.com/event/10791370	rdf:type	http://schema.org/Event	http://madrid.events.com/graph/events
6	http://madrid.events.com/event/11234187	rdf:type	http://schema.org/Event	http://madrid.events.com/graph/events

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Application

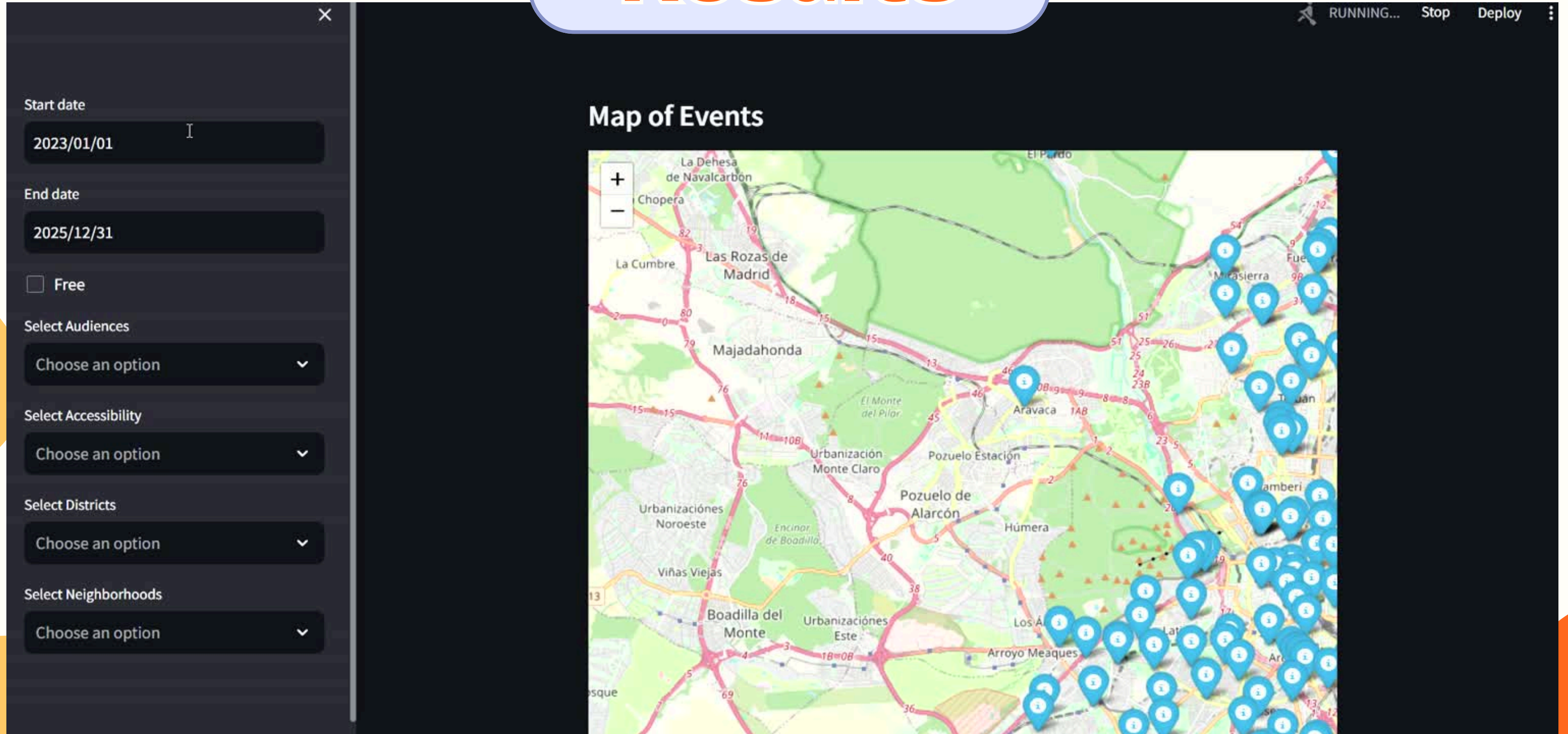
python, Streamlit



```
import streamlit as st
from SPARQLWrapper import SPARQLWrapper, JSON
import pandas as pd
import folium
from streamlit_folium import st_folium
```

```
endpoint_url = "http://localhost:7201/repositories/madrid_events"
sparql = SPARQLWrapper(endpoint_url)
sparql.setReturnFormat(JSON)
```

Results



Conclusion and Future Work

User-friendly application for accessing cultural events

Real-time visualization on a city map, allowing users to view, filter, and customize their search based on interests

- Add statistics
- Better visualization
- Link additional sources
- Mobile version
- Personalized event recommendations and alerts



THANK YOU!